# EEG reactivity in the prognosis of severe head injury

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**Article abstract**—We compared reactivity of EEG to external stimuli—an easily and quickly available measure—with the central conduction time (CCT) of the somatosensory evoked potentials, currently the most-used electrophysiologic method to predict outcome in severe head injury (SHI), and with the initial Glasgow Coma Scale (GCS) score. In 50 patients, comatose subsequent to SHI, we measured EEG reactivity and CCT within 48 to 72 hours and compared them with the outcome after 1.5 years. Using discriminant analysis, EEG reactivity correctly classified 92%, CCT classified 82%, and both measures together classified 98% of the patients into globally good or bad outcome groups. GCS allowed a correct classification in only 72% and, combined with either of the two electrophysiologic measures, did not further increase predictability. EEG reactivity is an excellent long-term global outcome predictor in SHI, superior to CCT and GCS. When the two electrophysiologic measures are combined, a prognostic accuracy is achieved that is better than that of any other reported method.

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Since EEG is influenced by drugs, particularly by high-dose barbiturates, its value for the prognostic evaluation of patients with severe head injury (SHI) is considered limited. Although its easy availability and the relatively short recording time make it an ideal tool even in smaller and less specialized hospital settings, somatosensory evoked potentials (SEPs) are preferred.[1-7] Although patients who are comatose subsequent to SHI are still frequently treated with sedatives and relaxants, the initial use of high-dose barbiturates is rather rare. This prompted us to reconsider the prognostic value of EEG, in particular its reactivity to external stimuli, as a prognostic variable in SHI.[8,9]

**Methods.** EEG and SEP recording was performed in 158 consecutively admitted patients, comatose subsequent to severe, non-missile head injury. The selection criteria for this study were age between 14 and 55 years, no known drug abuse, no history of psychiatric disease or of prior cerebral damage, pretraumatic knowledge of German, and a domicile within 100 km, in order to allow follow-up examination. The 50 patients fulfilling these criteria had a mean age of 30 years and a mean coma duration of 13 days (range, 2 to 60 days). Twenty-two patients were admitted with a Glasgow Coma Scale (GCS) score of 3 to 4, 16 with a GCS score of 5 to 6, and 12 with a GCS score of 7 to 8 (six had a short lucid interval immediately after the trauma). EEG and SEPs were recorded during coma within the first 48 to 72 hours after trauma. At the time of recording, all patients were sedated with midazolam; none received barbiturates.

*EEG.* EEG was recorded according to the 10-20 System in a conventional way. Reactivity was assessed using external auditory (loud click) and pain (nasal septum) stimulation. Analysis of EEG reactivity was done by visual inspection of the paper recording by two independent raters who had not examined the patients clinically. The analysis was performed immediately after the recording; the raters thus had no knowledge of the final outcome. Reactivity was rated according to one of four categories: occurrence of slow waves (figure 1A), flattening of the tracing (figure 1B), no reactivity (figure 1C), and doubtful. Ratings were considered concordant only if both raters agreed. In every instance of discordance, reactivity was considered doubtful. Thus, 24% of the patients reacted with slow waves, 30% reacted with a flattening of the tracing, in 18% reactivity was considered doubtful, and in 28% reactivity was absent.

*Somatosensory evoked potentials.* Stimulation was delivered by a bipolar surface electrode on the median nerve at the wrist to produce minimum thumb twitch. Impulse duration was 0.2 msec, stimulus rate 3/sec, and intensity 4 to 10 mA. The filter setting was 3 to 3,000 Hz. Recording electrodes were placed at Erb's point, at C2, and over the contralateral and ipsilateral parietal scalp referenced to linked earlobes. Each examination included three series of 1,024 trials on both sides. We evaluated the central conduction time (CCT), defined as the latency difference between N14 and the parietal N20. CCT was considered pathologic when prolonged beyond 2.5 SD of the normal mean value obtained from 20 normal subjects.[10]

*Clinical examinations and outcome estimation.* Ten patients died within 1 month after trauma. Eighteen months after trauma, the remaining 40 patients underwent a complete neurologic examination (always performed by E.G.) and mental status examination (always

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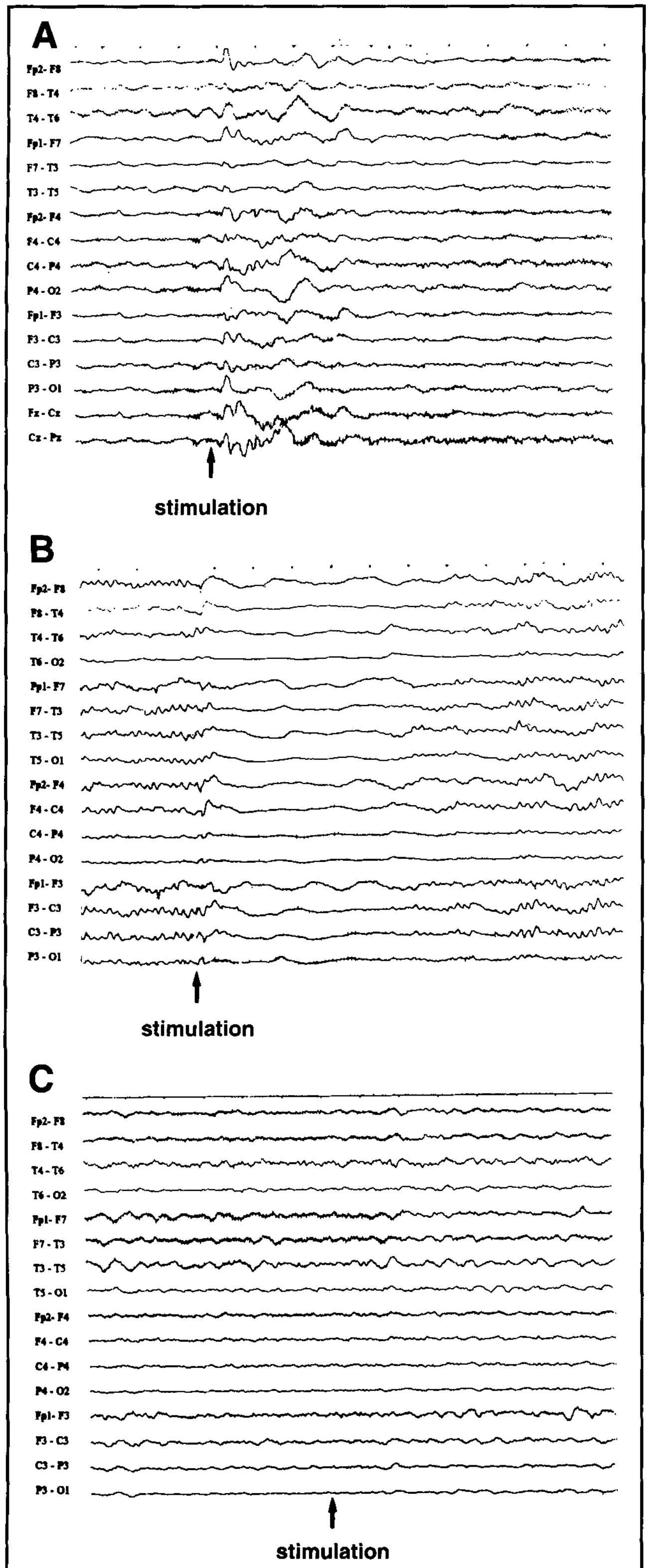


*Figure 1. Examples of EEG reactivity to external stimuli. (A) Occurrence of slow waves, (B) flattening of the tracing, and (C) no reactivity.*

performed by A.G.). Based on the neurologic and the mental status findings obtained 1.5 years after trauma, the patients were assigned either to a group with good outcome (Glasgow Outcome Score [GOS][11]: good recovery [GR] or moderate disability [MD]) or to one with bad outcome (severe disability, vegetative state, or death).

**Table 1. Clinical outcome after 18 months in relation to EEG reactivity, central conduction time, and Glasgow Coma Scale**

| Variables | Good outcome | | Bad outcome | | |
|---|---|---|---|---|---|
| | GR | MD | SD | VS | D |
| EEG reactivity | | | | | |
| Slow waves | 11 | 1 | — | — | — |
| Flattening | 7 | 7 | 1 | — | — |
| Doubtful | 3 | 4 | 1 | — | 1 |
| Absent | — | 1 | 3 | 1 | 9 |
| Central conduction time | | | | | |
| n n | 18 | 3 | 2 | 1 | 1 |
| n p | 3 | 4 | — | — | — |
| p p | — | 2 | — | — | — |
| n a | — | 3 | — | — | — |
| p a | — | 1 | 2 | — | — |
| a a | — | — | 1 | — | 9 |
| Glasgow Coma Scale | | | | | |
| 7-8 | 11 | 1 | — | — | — |
| 5-6 | 7 | 5 | 2 | — | 2 |
| 3-4 | 3 | 7 | 3 | 1 | 8 |

GR Good recovery.
MD Moderately disabled.
SD Severe disability.
VS Vegetative state.
D Death.
n Normal.
p Pathologic.
a Absent.

*Statistical analysis.* Discriminant analysis (SPSS/PC+, V3.1) used the two outcome groups as the dependent variable and EEG reactivity, CCT, and GCS as predictors.

EEG reactivity could have one of four values: slow waves = 3, flattening = 2, doubtful = 1, or absent = 0. GCS could have one of three values: GCS 3 to 4 = 1, GCS 5 to 6 = 2, or GCS 7 to 8 = 3. For CCT, one of three possible values (0, 1, or 2) was entered into the analysis for each state (normal [n], pathologic [p], and absent [a]), adding both hemispheric sides of registration.

Five values were thus obtained for each patient. For example, one patient reacted with slow waves in the EEG, showed a prolonged CCT on one side and a normal CCT on the other side, and had a GCS score of 5. The coding in this case was EEG reactivity = 3; CCT n = 1, p = 1, a = 0; GCS = 2.

**Results.** Thirty-four patients had a good and 16 patients a bad outcome. The individual GOS scores in relation to the initial EEG reactivity, CCT, and GCS group are displayed in table 1.

*EEG reactivity.* Whereas all but one patient with preserved EEG reactivity (96%) had a globally good outcome, 93% of the patients in whom EEG reactivity was absent had a bad outcome. True good recovery (GOS = GR) was found in 90% of the patients who reacted with slow waves but only in 50% of the patients who reacted with a flattening of the EEG tracing; the other 50% remained moderately disabled (GOS = MD). Although presence or absence of EEG reactivity best differentiated the good from the bad prognosis, it is of interest that 78% of those patients in whom EEG reactivity was doubtful had a good outcome.

*CCT.* Eighty-eight percent of the patients with bilaterally present cortical response and normal conduction on either side had a globally good outcome. However, true GR was found in 84% of the patients with bilaterally normal conduction and in only 43% of those with unilaterally normal conduction. None of the patients in whom conduction was prolonged bilaterally or the cortical response lacking unilaterally had GR, and 25% of these patients had a bad outcome. Lack of cortical response bilaterally was associated with a high mortality (90%); the only patient who survived remained severely disabled. This was the only patient in this group in whom EEG reactivity was preserved.

*GCS.* Whereas all the patients with a GCS score of 7 to 8 had a good outcome, in the group with GCS scores of 5 to 6 it was 75% and in the group with GCS scores of 3 to 4 it was 45%.

Figure 2A shows the EEG categories and figure 2B the CCT categories for each of the groups by initial coma scores (3 to 4, 5 to 6, and 7 to 8). Whereas EEG reactivity was absent in 55% of the patients admitted with a GCS score of 3 to 4, only 12.5% of the patients admitted with a GCS score of 5 to 6 and none of the patients with a GCS score of 7 to 8 showed nonreactive EEG tracings (figure 2A).

Bilaterally absent cortical SEPs were found only in the groups with GCS scores of 3 to 4 (32%) and 5 to 6 (19%), whereas all patients admitted with a GCS score of 7 to 8 had bilaterally normal SEPs (figure 2B).

*EEG reactivity, SEPs, and GCS as outcome predictors.* The results of discriminant analysis are displayed in table 2. When discriminant analysis used EEG reactivity alone as a predictor, it classified 92% of the patients correctly into a good- or bad-outcome group after 1.5 years; when using only the CCT of the SEPs it classified 82%; and a combined approach classified 98% correctly. GCS alone classified 72% correctly, GCS together with CCT 84%, GCS with EEG reactivity 94%, and GCS with both CCT and EEG reactivity 98%.

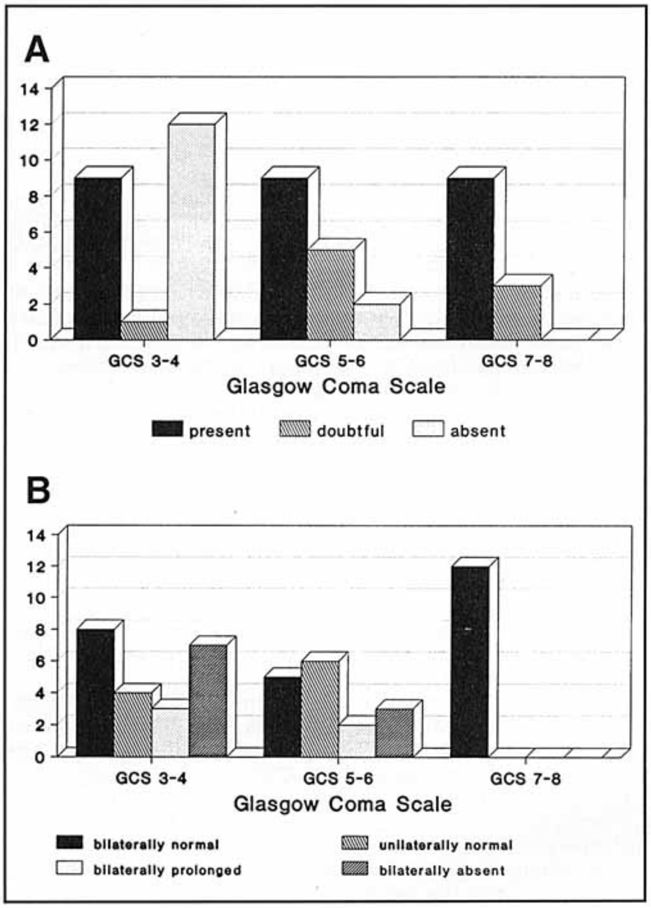


*Figure 2. (A) The number of patients in the respective EEG reactivity categories (present, doubtful, absent) for each of the initial Glasgow Coma Scale (GCS) groups (3 to 4, 5 to 6, 7 to 8). (B) The number of patients in the respective central conduction time categories—bilaterally normal, unilaterally normal (other side prolonged or absent), bilaterally prolonged, bilaterally absent—for each of the initial GCS groups (3 to 4, 5 to 6, 7 to 8).*

**Discussion.** EEG reactivity to external stimulation depends upon the intactness of anatomic pathways different from the lemnisco-thalamo-parietal axis known to carry somatosensory evoked responses. This was observed in early EEG studies. Moruzzi and Magoun[12] demonstrated in the cat that EEG reactivity depended on an intact reticular brainstem formation, and was mediated by a diffuse thalamocortical system and by an extrathalamic path through the internal capsule from the subthalamus and hypothalamus. Foltz and Schmidt[13] found dissociated reactivity of sensory evoked potentials recorded in the brainstem reticu-

**Table 2. Results of discriminant analysis**

| Outcome reached | Outcome predicted: EEG reactivity Good | EEG reactivity Bad | CCT Good | CCT Bad | EEG reactivity + CCT Good | EEG reactivity + CCT Bad | GCS Good | GCS Bad |
|---|---|---|---|---|---|---|---|---|
| Good | 33 | 1 | 31 | 3 | 34 | 0 | 24 | 10 |
| Bad | 3 | 13 | 6 | 10 | 1 | 15 | 4 | 12 |
| % Correctly classified | 92% | | 82% | | 98% | | 72% | |

CCT Central conduction time.
GCS Glasgow Coma Scale.

lar formation and in the medial lemniscus of monkeys immediately after concussion. Fischgold and Bounes[14] performed studies on EEG reactivity in patients with traumatic coma, and later[15,16] studied reactivity and periodicity of EEG in relation to cardiorespiratory function in coma of different etiologies. They found it similar to the potentials of the diffuse projection system described by Jasper[17] and conjectured that the absence of EEG reactivity represents a momentary deafferentation of the cerebrum as a sequela of a more or less reversible lesion. However, they did not think that EEG reactivity had prognostic value. Subsequent investigators were concerned with the prognostic value of the EEG in coma, but concentrated on EEG variables such as the frequency of the background activity[18,19] or the occurrence of different sleep patterns[8,20] rather than on EEG reactivity. In the last 15 years, SEPs became the method of choice for outcome estimation because of their resistance to sedative drugs. Because EEG reactivity and SEPs depend on different anatomic structures, thus possibly providing additional prognostic information, we evaluated EEG reactivity and compared its prognostic value for long-term outcome after 1.5 years with those of the two most frequently used prognostic measures, the GCS and CCT.

We were able to demonstrate that EEG reactivity is an excellent long-term global outcome predictor, classifying 92% of the patients correctly into either a good- or bad-outcome group. Its predictive power was even better than that of the CCT (82% correct classifications), hitherto the most frequently used electrophysiologic outcome predictor. When combining both measures, the prognostic accuracy increases to 98%, which is above the accuracy of any other reported method. GCS alone classified only 72% of patients correctly and was thus inferior to EEG as well as to CCT. Combining the initial coma score with either EEG or CCT scores did not improve the predictive power of the electrophysiologic measures substantially. Compared with both EEG and CCT, GCS gives too pessimistic a prediction, as 45% of the patients with an initial GCS score of 3 to 4 showed a good outcome, 40% normal EEG reactivity, and 36% bilaterally normal CCT. On the other hand, all but one patient in whom EEG reactivity was absent and all patients with missing cortical responses bilaterally showed a bad outcome. Bilaterally missing cortical responses were, moreover, associated with a high mortality (90%).

Our results show (1) EEG reactivity alone is an excellent predictor for the long-term global outcome in SHI and may well be an alternative to SEPs for intensive care units without evoked potential facilities. (2) The combination of the two measures increases predictive power, implying the involvement of different anatomic structures, as already suggested in early animal studies,[12-17] EEG reactivity primarily measuring the function of the reticular activating formation and SEPs that of the medial lemniscus. (3) GCS as a predictor in traumatic coma is inferior to both EEG reactivity and CCT, and its inclusion does not further increase predictability.

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